ALK fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ALK gene fusions from the msk_impact_50k_2026 study, covering 272 fusion events. 226/272 fusions (83.1%) were in-frame. 261/272 fusions (96.0%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test (p=0.00191661, statistically significant at alpha=0.05). No literature benchmark is configured for ALK.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 272 analyzed fusion events, identifying 32 distinct fusion partner genes. The most frequent partner was EML4, observed in 221 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 222/226 (98.2%) of in-frame fusions retained the domain, p=0.00191661 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 222/226 (98.2%) of in-frame fusions disrupted the domain, p=0.00768446 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.90099.

Cutpoint detection

Cutpoint detection scanned 271 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 972 aa (permutation-corrected p=0.00990099, statistically significant at alpha=0.05). This is 12 aa from the nearest configured domain boundary at 960 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=261) and "not_retained" (n=10). For Frame_status, retained: 222/227 (97.8%, 95% CI 94.9%-99.1%); not_retained: 4/5 (80.0%, 95% CI 37.6%-96.4%). Welch's t-test comparing Tumor_variant_count between groups gave means of 65.6139 vs 31.8, p=0.00432792 (statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (32 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: ALK has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EML4	221	0.8125	0.20043213737826426	0.0045279981461548964	0.8865878486549185	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.49628612459616434	1
LRP1B	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	1.0	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2706508884743972	2
EMILIN1	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.9559412550066756	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26404207672539853	3
CLTC	2	0.007352941176470588	0.20043213737826426	0.0045279981461548964	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2615402114933244	4
FN1	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2604372703168538	5
NOTCH1	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2604372703168538	6
NRP2	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2604372703168538	7
STRN	7	0.02573529411764705 8	0.20043213737826426	0.0045279981461548964	0.8865153538050734	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26024583860398176	8
KIF5B	5	0.01838235294117647	0.20043213737826426	0.0045279981461548964	0.8945260347129506	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2592415583872222	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PLEKHA7	1	0.003676470588235294	0.20043213737826426	0.004527998146154896	0.9158878504672897	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25803406604449064	10
SPTBN1	4	0.014705882352941176	0.20043213737826426	0.004527998146154896	0.8895193591455274	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2573876158756381	11
RANBP2	3	0.011029411764705883	0.20043213737826426	0.004527998146154896	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25563380687540244	12
PICALM	3	0.011029411764705883	0.20043213737826426	0.004527998146154896	0.8811748998664887	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2550330058073117	13
SQSTM1	2	0.007352941176470588	0.20043213737826426	0.004527998146154896	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25453086569893185	14
TPM1	2	0.007352941176470588	0.20043213737826426	0.004527998146154896	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25453086569893185	15
CTNNA1	1	0.003676470588235294	0.20043213737826426	0.004527998146154896	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	16
DCTN1	1	0.003676470588235294	0.20043213737826426	0.004527998146154896	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	17
DIAPH2	1	0.003676470588235294	0.20043213737826426	0.004527998146154896	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	18
DYSF	1	0.003676470588235294	0.20043213737826426	0.004527998146154896	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
GCC2	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	20
HMBOX1	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	21
PLEKHH2	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	22
PPP1CB	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	23
TFG	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	24
TRAF3	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	25
TTC27	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	26
VCL	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	27
ZFPM2	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25342792452246127	28
H6PD	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.7503337783711616	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23320095523007145	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CTSE	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.39118825100133514	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1793291261245975	30
SOS1	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	None	0.6853431332412958	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.14194222173458496	31
ASAP2	1	0.003676470588235294	0.20043213737826426	0.0045279981461548964	0.0	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12065088847439721	32

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
223	0	261	1	9	0.0	0.03690036900369004	1.4329692908744058
516	0	261	2	8	0.0	0.001230012300123001	2.9100905455940684
785	0	261	3	7	0.0	3.658029145347215e-05	4.436752838604533
939	2	259	3	7	0.018018018018018018	0.0003515999093716071	3.4539512455419974
972	2	259	4	6	0.011583011583011582	1.3820564716971664e-05	4.8594742110534295
1023	10	251	4	6	0.05976095617529881	0.0007961806348620647	3.098988389769786
1028	11	250	4	6	0.066	0.0010657190672512412	2.9723572639982643
1032	12	249	4	6	0.07228915662650602	0.0013947454739763106	2.8555050392241372
1035	13	248	4	6	0.07862903225806452	0.0017901726849544938	2.747105073765603
1040	14	247	4	6	0.08502024291497975	0.002258991897970247	2.646085326703862
1042	16	245	4	6	0.09795918367346938	0.003444927806265043	2.4628198749665864
1045	18	243	4	6	0.1111111111111111	0.005008607520231548	2.300282998750876
1047	19	242	4	6	0.11776859504132231	0.005949364209574388	2.225529443519691
1051	20	241	4	6	0.12448132780082988	0.007005060123461933	2.154588132872754
1052	22	239	4	6	0.13807531380753138	0.00948736346530418	2.0228544610771384
1058	254	7	10	0	0.0	1.0	-0.0
1059	255	6	10	0	0.0	1.0	-0.0
1061	258	3	10	0	0.0	1.0	-0.0
1064	259	2	10	0	0.0	1.0	-0.0
1070	260	1	10	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

222	40
4	5

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf00629	PF00629	271	9	0	262	0.0
auto_disruption_pf12810	PF12810	271	7	3	261	0.0016115098351940458

domain_retention tables

frame_domain_contingency_table

222	39
4	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	271	261	0	10	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
20	254	272	0.9338235294117647

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ASAP2	1
CLTC	2

Partner_gene	Event_count
CTNNA1	1
CTSE	1
DCTN1	1
DIAPH2	1
DYSF	1
EMILIN1	1
EML4	221
FN1	1
GCC2	1
H6PD	1
HMBOX1	1
KIF5B	5
LRP1B	1
NOTCH1	1
NRP2	1
PICALM	3
PLEKHA7	1
PLEKHH2	1
PPP1CB	1
RANBP2	3
SOS1	1
SPTBN1	4
SQSTM1	2
STRN	7
TFG	1
TPM1	2
TRAF3	1
TTC27	1
VCL	1
ZFPM2	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024679-T01-IM6-39	P-0024679-T01-IM6		ALK-CTSE	CTSE	unknown	five_prime	29543594	7	516	True	lost	0.0	False	PF00629	Protein kinase domain; PF12810		ALK (NM_004304) - CTSE(NM_001910) Rearrangement: t(1;2)(q32.1;p23.3)(chr1:g.206318648; chr2:g.29543594) | Antisense Fusion	NA
EVT-P-0020387-T01-IM6-40	P-0020387-T01-IM6		ALK-DCTN1	DCTN1	in-frame	three_prime	29446896	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		DCTN1 (NM_004082) - ALK (NM_004304) rearrangement: c.3196+238:DCTN1_c.3173-502:ALK del | Protein Fusion: in frame {DCTN1:ALK}	NA
EVT-P-0006776-T01-IM5-44	P-0006776-T01-IM5		ALK-DYSF	DYSF	unknown	five_prime	29447418	19	1058	True	lost	0.0	False	PF00629; PF12810	Protein kinase domain		ALK (NM_004304) - DYSF (NM_001130987) rearrangement: c.3172+909:ALK_c.3402+7348:DYSF dup | Antisense fusion	NA
EVT-P-0031827-T02-IM6-45	P-0031827-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446646	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-639:EML4_c.3173-252:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028287-T01-IM6-46	P-0028287-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446661	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-2623:EML4_c.3173-267:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0009301-T01-IM5-47	P-0009301-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446651	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.668-6693:EML4_c.3173-257:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009301-T02-IM6-48	P-0009301-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446651	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.668-6693:EML4_c.3173-257:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009339-T01-IM5-49	P-0009339-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447889	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-6870:EML4_c.3172+438:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009339-T02-IM5-50	P-0009339-T02-IM5		ALK-EML4	EML4	in-frame	three_prime	29447889	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-6870:EML4_c.3172+438:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009343-T01-IM5-51	P-0009343-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446521	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-1093:EML4_c.3173-127:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009353-T01-IM5-52	P-0009353-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446465	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.667+2407:EML4_c.3173-71:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0031406-T01-IM6-53	P-0031406-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446425	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 with ALK exons 20-29): c.668-2889: EML4_c.3173-31:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032170-T01-IM6-54	P-0032170-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447695	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1822:EML4_c.3172+632:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0010041-T01-IM5-55	P-0010041-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447942	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fuses with ALK exons 20-29) : c.701-6977:EML4_c.3172+385:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0008332-T01-IM5-56	P-0008332-T01-IM5		ALK-EML4	EML4	unknown	five_prime	29446912	19	1058	True	lost	0.0	False	PF00629; PF12810	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) rearrangement : c.1489+828:EML4_c.3173-518:ALKdup | Antisense fusion	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010177-T01-IM5-57	P-0010177-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447358	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29): c.1490-2649:EML4_c.3173-964:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0007705-T01-IM5-58	P-0007705-T01-IM5		ALK-EML4	EML4	unknown	three_prime	29447417	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1594: EML4_c.3172+910:ALKinv | Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0010978-T01-IM5-59	P-0010978-T01-IM5		ALK-EML4	EML4	out-of-frame	three_prime	29446414	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv | Protein fusion: out of frame (EML4-ALK)	NA
EVT-P-0010978-T02-IM5-60	P-0010978-T02-IM5		ALK-EML4	EML4	out-of-frame	three_prime	29446414	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv | Protein fusion: out of frame (EML4-ALK)	NA
EVT-P-0011018-T01-IM5-61	P-0011018-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29448314	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011203-T01-IM5-62	P-0011203-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447105	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011203-T02-IM5-63	P-0011203-T02-IM5		ALK-EML4	EML4	in-frame	three_prime	29447096	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6805:EML4_c.3173-702:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011203-T03-IM6-64	P-0011203-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29447055	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 5 fused to ALK exon 20): c.494-6851:EML4_c.3173-661:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012679-T01-IM5-65	P-0012679-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447466	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1489+2812:EML4_c.3172+861:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0012778-T01-IM5-66	P-0012778-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447442	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1320:EML4_c.3172+885:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0012778-T02-IM6-67	P-0012778-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29447442	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+1320:EML4_c.3172+885:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0007211-T01-IM5-68	P-0007211-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447489	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.667+165:EML4_c.3172+838:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0007125-T01-IM5-69	P-0007125-T01-IM5		ALK-EML4	EML4	unknown	three_prime	29446597	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-14 with ALK exons 20-29) : c.1328:EML4_c.3173-203:ALKinv | Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0006950-T02-IM6-70	P-0006950-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29447384	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons20-29): c.1489+2302:EML4_c.3172+943:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0013202-T01-IM5-71	P-0013202-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447178	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+2565:EML4_c.3173-784:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006950-T01-IM5-72	P-0006950-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447384	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1489+2302:EML4_c.3172+943:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013453-T01-IM5-73	P-0013453-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446597	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+551:EML4_c.3173-203:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013864-T01-IM5-74	P-0013864-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29448209	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1489+658:EML4_c.3172+118:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013869-T01-IM5-75	P-0013869-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446712	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1490-187:EML4_c.3173-318:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014270-T01-IM6-76	P-0014270-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446679	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294:EML4_c.3173-285:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014310-T01-IM6-77	P-0014310-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447544	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20- 29): c.667+6128:EML4_c.3172+783:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006370-T01-IM5-78	P-0006370-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29448124	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1490-2147:EML4_c.3172+203:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014385-T01-IM6-79	P-0014385-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447676	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1890:EML4_c.3172+651:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006313-T01-IM5-80	P-0006313-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447966	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) -ALK (NM_004304) Fusion (EML4 exons 1 to 18 fused with exons 20 to 29 of ALK) | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0030590-T01-IM6-81	P-0030590-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447704	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.494-5628:EML4_c.3173-350:ALKinv. | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0000474-T01-IM3-82	P-0000474-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447002	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0000557-T01-IM3-83	P-0000557-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29446543	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0000732-T01-IM3-84	P-0000732-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447537	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015197-T01-IM6-85	P-0015197-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448009	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6 fused to ALK exon 20) : c.668-2108:EML4_c.3172+318:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015434-T02-IM6-86	P-0015434-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29448197	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 13 fused to ALK exon20): c.1489+63:EML4_c.3172+130:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005321-T01-IM5-87	P-0005321-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29448250	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 13 fused with ALK exon 20) : c.2068+156:EML4_c.3173-693:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0029971-T01-IM6-88	P-0029971-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447807	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-119:EML4_c.3172+520:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015921-T03-IM6-89	P-0015921-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446809	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29): c.667+516:EML4_c.3173-415:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015961-T01-IM6-90	P-0015961-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6fused to ALK exon 20): c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015961-T02-IM6-91	P-0015961-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015961-T03-IM6-92	P-0015961-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031827-T01-IM6-93	P-0031827-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446700	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-673:EML4_c.3173-306:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027119-T01-IM6-94	P-0027119-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446711	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused to ALK exons 20-29): c.1490-837:EML4_c.3173-317:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029750-T01-IM6-95	P-0029750-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446660	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2574:EML4_c.3173-266:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0029550-T01-IM6-96	P-0029550-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447863	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-721:EML4_c.3172+464:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0016280-T01-IM6-97	P-0016280-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447905	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) rearrangement: c.1316-2732:EML4_c.3172+422:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0016580-T01-IM6-98	P-0016580-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447579	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+2257:EML4_c.3172+748:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0005305-T01-IM5-99	P-0005305-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447764	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-5757: EML4_c.3172+563:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0016960-T01-IM6-100	P-0016960-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447838	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7453:EML4_c.3172+489:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017592-T01-IM5-101	P-0017592-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447548	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 1-5 fused with ALK exon 20-29) : c.494-5963:EML4_c.3172+779:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029400-T01-IM6-102	P-0029400-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447668	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1315+641:EML4_c.3172+659:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0004218-T01-IM5-103	P-0004218-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447087	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 20 fused with ALK exon 20) : c.2068+156:EML4_c.3173-693:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0017978-T01-IM6-104	P-0017978-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017978-T02-IM6-105	P-0017978-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017978-T03-IM6-106	P-0017978-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018242-T01-IM6-107	P-0018242-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448007	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused in-frame with ALK exons 20-29): c.668-7560:EML4_c.3172+320:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018280-T01-IM6-108	P-0018280-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447187	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 6 fused with ALK exon 20): c.494-6792:EML4_c.3173-793:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003913-T01-IM3-109	P-0003913-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447211	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0029174-T01-IM6-110	P-0029174-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446656	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-19 fused with ALK exons 20-29): c.2242+168:EML4_c.3173-262:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018969-T01-IM6-112	P-0018969-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29448416	19	1028	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-760:EML4_c.3083:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019491-T01-IM6-113	P-0019491-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446596	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1490-486:EML4_c.3173-202:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019513-T01-IM6-114	P-0019513-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29448403	19	1032	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 19-29): c.1489+2278:EML4_c.3096:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019533-T01-IM6-115	P-0019533-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446494	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+6837:EML4_c.3173-100:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0009102-T01-IM5-116	P-0009102-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446513	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.1490-244:EML4_c.3173-119:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028901-T01-IM6-117	P-0028901-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446607	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-844:EML4_c.3173-213:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019786-T01-IM6-118	P-0019786-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447575	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+82:EML4_c.3172+752:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019972-T01-IM6-119	P-0019972-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448101	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused to ALK exon 20-29) : c.493+1179:EML4_c.3172+226:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020124-T01-IM6-120	P-0020124-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447955	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020138-T01-IM6-121	P-0020138-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447907	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-3016:EML4_c.3172+420:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003383-T01-IM5-122	P-0003383-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447572	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0020408-T01-IM6-123	P-0020408-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446779	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1227:EML4_c.3173-385:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003323-T01-IM5-124	P-0003323-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29447530	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003297-T01-IM5-125	P-0003297-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29448259	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0020584-T01-IM6-126	P-0020584-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447048	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exon 20-29) : c.209-4846:EML4_c.3173-654:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020780-T01-IM6-127	P-0020780-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447084	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1 - 13 fused to ALK exons 20 - 29): c.1489+1278:EML4_c.3173-690:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020825-T03-IM6-128	P-0020825-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446767	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29):c.208+4890:EML4_c.3173-373:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0021257-T01-IM6-129	P-0021257-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447777	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-17 fused with ALK exons 20-29): c.1883-596:EML4_c.3172+550:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003206-T01-IM5-130	P-0003206-T01-IM5		ALK-EML4	EML4	unknown	three_prime	29448365	19	1045	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0021495-T01-IM6-131	P-0021495-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448313	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+2355:EML4_c.3172+14:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002613-T01-IM3-132	P-0002613-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447199	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0002408-T02-IM5-133	P-0002408-T02-IM5		ALK-EML4	EML4	in-frame	three_prime	29448053	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fuses with ALK exons 20-29) : c.2057-115:EML4_c.3172+274:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0021693-T01-IM6-134	P-0021693-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447002	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-14 fused with ALK exons 20-29): c.1316-1509:EML4_c.3173-608:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022101-T01-IM6-135	P-0022101-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447759	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon18 fused with ALK exon 20): c.2057-510:EML4_c.2057-510:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022138-T01-IM6-136	P-0022138-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446584	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused with ALK exons 20-29): c.2333:EML4_c.3173-190:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0036247-T01-IM6-137	P-0036247-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447857	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+107:EML4_c.3172+470:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036106-T01-IM6-138	P-0036106-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446930	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028745-T01-IM6-139	P-0028745-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447509	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022398-T01-IM6-140	P-0022398-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448035	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-7853:EML4_c.3172+292:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002266-T01-IM3-141	P-0002266-T01-IM3		ALK-EML4	EML4	in-frame	five_prime	29447288	19	1058	True	lost	0.0	False	PF00629; PF12810	Protein kinase domain		ALK (NM_004304) and EML4 (NM_019063) reciprocal inversion (13026566bp): c.3173-889:ALK_c.208+1022:EML4inv | Protein fusion: in frame (ALK-EML4)	NA
EVT-P-0002266-T01-IM3-142	P-0002266-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447283	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0028541-T03-IM6-143	P-0028541-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion: c.208+4631:EML4_c.3173-528:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022926-T01-IM6-144	P-0022926-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446558	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+983:EML4_c.3173-164:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0035121-T01-IM6-145	P-0035121-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447383	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4857:EML4_c.3172+944:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002217-T02-IM3-146	P-0002217-T02-IM3		ALK-EML4	EML4	in-frame	three_prime	29446594	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0023368-T01-IM6-147	P-0023368-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446713	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29 including the kinase domain) : c.667+7004:EML4_c.3173-319:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002024-T01-IM3-148	P-0002024-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447133	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) inversion: c.2242+31:EML4_c.3173-739:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0023962-T01-IM6-149	P-0023962-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448820	19	1023	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775:EML4_c.3068-389:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0024136-T01-IM6-150	P-0024136-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448266	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-4498:EML4_c.3172+61:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0001811-T01-IM3-151	P-0001811-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29448205	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666:EML4_c.3172+122:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0028541-T02-IM6-152	P-0028541-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631:EML4_c.3173-528:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0024907-T01-IM6-153	P-0024907-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446877	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29): c.209-990:EML4_c.3173-483:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034569-T02-IM6-154	P-0034569-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29446514	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-217:EML4_c.3173-120:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034450-T01-IM6-155	P-0034450-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446354	20	1071	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001532-T01-IM3-156	P-0001532-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447553	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0028541-T01-IM6-157	P-0028541-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446922	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631:EML4_c.3173-528:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0026407-T01-IM6-158	P-0026407-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446392	20	1059	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.1981-2450:EML4_c.3175:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0026650-T02-IM6-159	P-0026650-T02-IM6		ALK-EML4	EML4	in-frame	five_prime	29447451	19	1058	True	lost	0.0	False	PF00629; PF12810	Protein kinase domain		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.667+7374:EML4_c.3172+876:ALKinv | Protein Fusion: in frame {ALK:EML4}	NA
EVT-P-0026657-T01-IM6-160	P-0026657-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29446952	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0026657-T03-IM6-161	P-0026657-T03-IM6		ALK-EML4	EML4	unknown	three_prime	29446952	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312: EML4_c.3173-558:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0015152-T02-IM6-162	P-0015152-T02-IM6		ALK-EML4	EML4	in-frame	three_prime	29447079	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused in-frame to ALK exons 20-29): c.2242+127:EML4_c.3173-685:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032862-T01-IM6-163	P-0032862-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447488	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) Fusion(EML4 exon 11 fused to ALK exon20): c.1489+671:EML4_c.3172+839:ALK | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027119-T03-IM6-164	P-0027119-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446711	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-837:EML4_c.3173-317:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032683-T01-IM6-165	P-0032683-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447763	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028339-T01-IM6-166	P-0028339-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447680	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+1245:EML4_c.3172+647:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032518-T01-IM6-167	P-0032518-T01-IM6		ALK-EML4	EML4	unknown	three_prime	29448359	19	1047	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+2607:EML4_c.3140:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0027853-T01-IM6-168	P-0027853-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447734	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-157:EML4_c.3172+593:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015110-T01-IM6-169	P-0015110-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29446518	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668-1726:EML4_c.3173-124:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003446-T01-IM5-170	P-0003446-T01-IM5		ALK-EML4	EML4	in-frame	three_prime	29446534	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004516-T02-IM5-171	P-0004516-T02-IM5		ALK-GCC2	GCC2	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		GCC2 (NM_181453) - ALK (NM_004304) rearrangement : c.4448-1230:GCC2_c.3173-132:ALKinv | Protein fusion: in frame (GCC2-ALK)	NA
EVT-P-0035712-T01-IM6-172	P-0035712-T01-IM6		ALK-H6PD	H6PD	unknown	five_prime	29462508	13	785	True	lost	0.0	False	PF00629	Protein kinase domain	PF12810	ALK (NM_004304) rearrangement: t(1;2)(p36.22;p23.2)(chr1:g.9314150::chr2:g.29462508) | Antisense Fusion	NA
EVT-P-0018541-T01-IM6-174	P-0018541-T01-IM6		ALK-LRP1B	LRP1B	in-frame	five_prime	29450289	17	972	True	lost	0.0	False	PF00629; PF12810	Protein kinase domain		ALK (NM_004304) - LRP1B (NM_018557) Rearrangement : c.2914+151:ALK_c.344-85984:LRP1Bdup | Protein Fusion: in frame {ALK:LRP1B}	NA
EVT-P-0027092-T01-IM6-176	P-0027092-T01-IM6		ALK-PLEKHH2	PLEKHH2	in-frame	three_prime	29446506	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		PLEKHH2 (NM_172069) - ALK (NM_004304) rearrangement:c.502+545:PLEKHH2_c.3173-112:ALKinv | Protein Fusion: in frame {PLEKHH2:ALK}	NA
EVT-P-0027363-T02-IM6-177	P-0027363-T02-IM6		ALK-RANBP2	RANBP2	in-frame	three_prime	29447364	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1556:RANBP2_c.3172+963:ALKinv | Protein Fusion: in frame {RANBP2:ALK}	NA
EVT-P-0068865-T02-IM7-178	P-0068865-T02-IM7		ALK-RANBP2	RANBP2	in-frame	five_prime	29447373	19	1058	True	lost	0.0	False	PF00629; PF12810	Protein kinase domain		ALK (NM_004304) - RANBP2 (NM_006267) rearrangement: c.3172+954:ALK_c.2602+1532:RANBP2inv | Protein Fusion: in frame {ALK:RANBP2}	NA
EVT-P-0008843-T01-IM5-179	P-0008843-T01-IM5		ALK-RANBP2	RANBP2	in-frame	three_prime	29446488	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1070:RANBP2_c.3173-94:ALKinv | Protein fusion: in frame (RANBP2-ALK)	NA
EVT-P-0015699-T01-IM6-181	P-0015699-T01-IM6		ALK-SPTBN1	SPTBN1	in-frame	three_prime	29447437	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		SPTBN1 (NM_003128) - ALK (NM_004304) fusion (SPTBN1 exon 7 fused to ALK exon 20): c.764-1028:SPTBN1_c.3172+890:ALKinv | Protein fusion: in frame (SPTBN1-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016193-T05-IM6-182	P-0016193-T05-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		STRN (NM_003162) - ALK (NM_004304) fusion (STRN exons 1-3 fused with ALK exons 20-29): c.412+4919:STRN_c.3172+1:ALKdel | Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0016193-T02-IM6-183	P-0016193-T02-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		ALK (NM_004304) - STRN (NM_003162) :c.3172+1:ALK_c.412+4919:STRNdel | Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0016193-T01-IM6-184	P-0016193-T01-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		ALK (NM_004304) - STRN (NM_003162) :c.3172+1:ALK_c.412+4919:STRNdel | Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0041438-T01-IM6-185	P-0041438-T01-IM6		ALK-TTC27	TTC27	out-of-frame	five_prime	29447817	19	1058	True	lost	0.0	False	PF00629; PF12810	Protein kinase domain		ALK (NM_004304) - TTC27 (NM_017735) rearrangement: c.3172+510:ALK_c.538-4165:TTC27inv | Protein Fusion: out of frame {ALK:TTC27}	NA
EVT-P-0006617-T01-IM5-187	P-0006617-T01-IM5		ASAP2-ALK	ASAP2	unknown	five_prime	30142753	1	223	True	lost	0.0	False		Protein kinase domain; PF00629; PF12810		ASAP2 (NM_003887) - ALK (NM_004304) rearrangement event : c.127-20148:ASAP2_c.667+106:ALKdel | Antisense fusion	NA
EVT-P-0062994-T02-IM7-188	P-0062994-T02-IM7		CLTC-ALK	CLTC	in-frame	three_prime	29449624	19	1023	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23.2;q23.1)(chr2:g.29449624::chr17:g.57768350) | Protein Fusion: in frame {CLTC:ALK}	NA
EVT-P-0051864-T01-IM6-189	P-0051864-T01-IM6		CLTC-ALK	CLTC	in-frame	three_prime	29448839	19	1023	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23.2;q23.1)(chr2:g.29448839::chr17:g.57770376) | Protein Fusion: in frame {CLTC:ALK}	NA
EVT-P-0025388-T01-IM6-190	P-0025388-T01-IM6		CTNNA1-ALK	CTNNA1	in-frame	three_prime	29446465	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		CTNNA1 (NM_001903) - ALK (NM_004304) rearrangment: t(2;5)(p23.2;q31.2)(chr2:g.29446465::chr5:g.138259019) | Protein Fusion: in frame {CTNNA1:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051068-T01-IM6-191	P-0051068-T01-IM6		DIAPH2-ALK	DIAPH2	in-frame	three_prime	29447566	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		DIAPH2 (NM_006729) - ALK (NM_004304) fusion: t(2;X)(p23.2;q21.33)(chr2:g.29447566::chrX:g.96713620) | Protein Fusion: in frame {DIAPH2:ALK}	NA
EVT-P-0005759-T02-IM6-192	P-0005759-T02-IM6		EMILIN1-ALK	EMILIN1	in-frame	three_prime	29450741	17	939	True	retained	1.0	False	Protein kinase domain	PF00629	PF12810	EMILIN1 (NM_007046) - ALK (NM_004304) fusion: c.2575+44:EMILIN1_c.2816-203:ALKinv | Protein Fusion: in frame {EMILIN1:ALK}	NA
EVT-P-0046083-T01-IM6-193	P-0046083-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446665	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+986:EML4_c.3173-271:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049375-T01-IM6-194	P-0049375-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446732	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015646-T03-IM6-195	P-0015646-T03-IM6		EML4-ALK	EML4	in-frame	three_prime	29448028	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.209-3909:EML4_c.3172+299:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014385-T03-IM6-196	P-0014385-T03-IM6		EML4-ALK	EML4	in-frame	three_prime	29447676	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027119-T04-IM7-197	P-0027119-T04-IM7		EML4-ALK	EML4	in-frame	three_prime	29446711	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-837:EML4_c.3173-317:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0057391-T02-IM6-198	P-0057391-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29447092	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3018:EML4_c.3173-698:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034334-T01-IM6-199	P-0034334-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446570	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4909:EML4_c.3173-176:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025095-T01-IM6-200	P-0025095-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447321	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20 -29): c.2243-278:EML4_c.3173-927:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0025040-T01-IM6-201	P-0025040-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447152	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20) : c.1316-1940:EML4_c.3173-758:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0007705-T02-IM6-202	P-0007705-T02-IM6		EML4-ALK	EML4	unknown	three_prime	29447417	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0023375-T02-IM7-203	P-0023375-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447854	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2728:EML4_c.3172+473:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0035085-T03-IM6-204	P-0035085-T03-IM6		EML4-ALK	EML4	in-frame	three_prime	29447079	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7361:EML4_c.3173-685:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0001766-T04-IM6-205	P-0001766-T04-IM6		EML4-ALK	EML4	in-frame	three_prime	29448767	19	1023	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3962:EML4_c.3068-336:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0068467-T02-IM7-206	P-0068467-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29448295	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1566:EML4_c.3172+32:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022184-T04-IM6-207	P-0022184-T04-IM6		EML4-ALK	EML4	in-frame	three_prime	29447035	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1386:EML4_c.3173-641:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036106-T03-IM7-208	P-0036106-T03-IM7		EML4-ALK	EML4	in-frame	three_prime	29446930	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-536:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022184-T01-IM6-209	P-0022184-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447035	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1315+1386:EML4_c.3173-641:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037404-T01-IM6-210	P-0037404-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447872	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4963:EML4_c.3172+455:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037491-T01-IM6-211	P-0037491-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446786	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon12 fused with ALK exon20): c.1490-1120:EML4_c.3173-392:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037668-T01-IM6-212	P-0037668-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447631	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2742:EML4_c.3172+696:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037925-T01-IM6-213	P-0037925-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446522	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4634:EML4_c.3173-128:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037982-T01-IM6-214	P-0037982-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446479	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-138:EML4_c.3173-85:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0038201-T03-IM6-215	P-0038201-T03-IM6		EML4-ALK	EML4	in-frame	three_prime	29447748	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+989:EML4_c.3172+579:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0038772-T01-IM6-216	P-0038772-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446687	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2341+147:EML4_c.3173-293:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043860-T02-IM6-217	P-0043860-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446931	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-980:EML4_c.3173-537:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0067093-T02-IM7-218	P-0067093-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447303	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+1508:EML4_c.3173-909:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017592-T04-IM6-219	P-0017592-T04-IM6		EML4-ALK	EML4	in-frame	three_prime	29447548	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-5963:EML4_c.3172+779:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040261-T01-IM6-220	P-0040261-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447103	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1340:EML4_c.3173-709:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015961-T07-IM6-221	P-0015961-T07-IM6		EML4-ALK	EML4	in-frame	three_prime	29446526	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040553-T01-IM6-222	P-0040553-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447055	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+30:EML4_c.3173-661:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040574-T01-IM6-223	P-0040574-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447738	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-1589:EML4_c.3172+589:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040576-T01-IM6-224	P-0040576-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447321	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-821:EML4_c.3173-927:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040600-T01-IM6-225	P-0040600-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447688	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1795:EML4_c.3172+639:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032683-T02-IM7-226	P-0032683-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447763	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014385-T06-IM7-227	P-0014385-T06-IM7		EML4-ALK	EML4	in-frame	three_prime	29447676	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0067069-T01-IM7-228	P-0067069-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29448169	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2305:EML4_c.3172+158:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014385-T02-IM6-229	P-0014385-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29447676	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0041967-T01-IM6-230	P-0041967-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29448147	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+447:EML4_c.3172+180:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0041967-T02-IM7-231	P-0041967-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29448147	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+447:EML4_c.3172+180:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0067048-T02-IM7-232	P-0067048-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447391	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+2216:EML4_c.3172+936:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0066568-T02-IM7-233	P-0066568-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29448072	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+3428:EML4_c.3172+255:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014310-T02-IM6-234	P-0014310-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29447544	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+6128:EML4_c.3172+783:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043041-T01-IM6-235	P-0043041-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446597	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1782:EML4_c.3173-203:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013299-T02-IM6-236	P-0013299-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29447038	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) -ALK (NM_004304) Fusion(EML4 exon5 fused with ALK exon20) : c.667+1302:EML4_c.3173-644:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012913-T02-IM6-237	P-0012913-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446965	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.209-4133:EML4_c.3173-571:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043350-T01-IM6-238	P-0043350-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447513	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4017:EML4_c.3172+814:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043363-T01-IM6-239	P-0043363-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29448101	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2254:EML4_c.3172+226:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0043594-T01-IM6-240	P-0043594-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29448381	19	1040	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.339-2119:EML4_c.3118:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0069396-T02-IM7-241	P-0069396-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446944	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4308:EML4_c.3173-550:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069535-T01-IM7-242	P-0069535-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29447764	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069617-T01-IM7-243	P-0069617-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29448017	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0044432-T01-IM6-244	P-0044432-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29448344	19	1052	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2057-576:EML4_c.3155:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005743-T02-IM5-245	P-0005743-T02-IM5		EML4-ALK	EML4	in-frame	three_prime	29447669	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 with ALK exons 20-29) : c.2243-181:EML4_c.3172+658:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0045222-T03-IM7-246	P-0045222-T03-IM7		EML4-ALK	EML4	unknown	three_prime	29447583	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0045243-T01-IM6-247	P-0045243-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29448298	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0045341-T01-IM6-248	P-0045341-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447920	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0065469-T02-IM7-249	P-0065469-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446565	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0065210-T01-IM7-250	P-0065210-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29448076	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2012:EML4_c.3172+251:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012873-T03-IM7-251	P-0012873-T03-IM7		EML4-ALK	EML4	in-frame	three_prime	29446439	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0064716-T02-IM7-252	P-0064716-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447739	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3669:EML4_c.3172+588:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031368-T04-IM6-253	P-0031368-T04-IM6		EML4-ALK	EML4	in-frame	three_prime	29447191	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1154:EML4_c.3173-797:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045912-T02-IM6-254	P-0045912-T02-IM6		EML4-ALK	EML4	out-of-frame	three_prime	29446507	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+1011:EML4_c.3173-113:ALKinv | Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0031824-T01-IM6-255	P-0031824-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29450557	17	939	True	retained	1.0	False	Protein kinase domain	PF00629	PF12810	EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+2971:EML4_c.2816-19:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0046083-T02-IM7-256	P-0046083-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446665	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+986:EML4_c.3173-271:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0046594-T01-IM6-257	P-0046594-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446445	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+2217:EML4_c.3173-51:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040732-T02-IM6-258	P-0040732-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446636	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1832:EML4_c.3173-242:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0047355-T01-IM6-259	P-0047355-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0047355-T02-IM6-260	P-0047355-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446609	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0064342-T01-IM7-261	P-0064342-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29446534	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0063585-T01-IM7-262	P-0063585-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29446884	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012873-T02-IM6-263	P-0012873-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446439	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0062999-T02-IM7-264	P-0062999-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447091	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049318-T01-IM6-265	P-0049318-T01-IM6		EML4-ALK	EML4	unknown	three_prime	29446359	20	1070	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+2335:EML4_c.3208:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0046720-T01-IM6-266	P-0046720-T01-IM6		EML4-ALK	EML4	out-of-frame	three_prime	29446669	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1641+1:EML4_c.3173-275:ALKinv | Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0049375-T02-IM6-267	P-0049375-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29446732	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049511-T01-IM6-268	P-0049511-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446991	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0010177-T02-IM7-269	P-0010177-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447358	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031827-T03-IM6-270	P-0031827-T03-IM6		EML4-ALK	EML4	in-frame	three_prime	29446646	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0051231-T01-IM6-271	P-0051231-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446774	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0051231-T03-IM6-272	P-0051231-T03-IM6		EML4-ALK	EML4	in-frame	three_prime	29446774	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0062985-T02-IM7-273	P-0062985-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29447199	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4561:EML4_c.3173-805:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052035-T03-IM6-274	P-0052035-T03-IM6		EML4-ALK	EML4	in-frame	three_prime	29447656	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+79:EML4_c.3172+671:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0061297-T03-IM7-275	P-0061297-T03-IM7		EML4-ALK	EML4	unknown	three_prime	29448375	19	1042	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+931:EML4_c.3124:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0052297-T02-IM6-276	P-0052297-T02-IM6		EML4-ALK	EML4	in-frame	three_prime	29448255	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-161:EML4_c.3172+72:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0061297-T02-IM7-277	P-0061297-T02-IM7		EML4-ALK	EML4	unknown	three_prime	29448375	19	1042	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+931:EML4_c.3124:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0005786-T01-IM5-278	P-0005786-T01-IM5		EML4-ALK	EML4	in-frame	three_prime	29448239	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 with ALK exons 20-29) : c.2057-186:EML4_c.3172+88:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0060148-T01-IM7-279	P-0060148-T01-IM7		EML4-ALK	EML4	in-frame	three_prime	29447757	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2600:EML4_c.3172+570:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0059699-T03-IM7-280	P-0059699-T03-IM7		EML4-ALK	EML4	in-frame	three_prime	29447467	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1990:EML4_c.3172+860:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031827-T06-IM7-281	P-0031827-T06-IM7		EML4-ALK	EML4	in-frame	three_prime	29446646	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0054263-T01-IM6-282	P-0054263-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446557	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-210:EML4_c.3173-163:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059257-T02-IM7-283	P-0059257-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29448210	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-982:EML4_c.3172+117:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0054408-T01-IM6-284	P-0054408-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29446902	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.2243-262:EML4_c.3173-508:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0008332-T02-IM6-285	P-0008332-T02-IM6		EML4-ALK	EML4	unknown	three_prime	29446913	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+828:EML4_c.3173-519:ALKdup | Antisense Fusion	NA
EVT-P-0058599-T02-IM7-286	P-0058599-T02-IM7		EML4-ALK	EML4	in-frame	three_prime	29446415	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1345:EML4_c.3173-21:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0044042-T01-IM6-287	P-0044042-T01-IM6		EML4-ALK	EML4	in-frame	three_prime	29447727	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-1681:EML4_c.3172+600:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019560-T01-IM6-288	P-0019560-T01-IM6		EML4-ALK	EML4	out-of-frame	three_prime	29447952	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-16 with ALK exons 20-29): c.1968-2879:EML4_c.3172+375:ALKinv | Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0044526-T02-IM6-289	P-0044526-T02-IM6		FN1-ALK	FN1	in-frame	three_prime	29448533	19	1023	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		FN1 (NM_212482) - ALK (NM_004304) fusion: c.1819+332:FN1_c.3068-102:ALKdel | Protein Fusion: in frame {FN1:ALK}	NA
EVT-P-0030808-T01-IM6-290	P-0030808-T01-IM6		HMBOX1-ALK	HMBOX1	unknown	three_prime	29447713	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		HMBOX1 (NM_001135726) ALK (NM_004304) Rearrangement : t(2,8)(p23.3,p21.1)(chr2:g.29447713::chr8:g.28904937) | Protein Fusion: mid-exon {HMBOX1:ALK}	NA
EVT-P-0062804-T03-IM7-291	P-0062804-T03-IM7		KIF5B-ALK	KIF5B	in-frame	three_prime	29447791	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447791::chr10:g.32304833) | Protein Fusion: in frame {KIF5B:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005711-T02-IM6-292	P-0005711-T02-IM6		KIF5B-ALK	KIF5B	in-frame	three_prime	29449548	19	1023	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29449548::chr10:g.32304765) | Protein fusion: in frame (KIF5B-ALK)	NA
EVT-P-0008764-T01-IM5-293	P-0008764-T01-IM5		KIF5B-ALK	KIF5B	unknown	three_prime	29448327	19	1058	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.32305750) | Protein fusion: mid-exon (KIF5B-ALK)	NA
EVT-P-0017924-T01-IM6-294	P-0017924-T01-IM6		KIF5B-ALK	KIF5B	in-frame	three_prime	29447945	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447945::chr10:g.32305310) | Protein Fusion: in frame {KIF5B:ALK}	NA
EVT-P-0008764-T02-IM6-295	P-0008764-T02-IM6		KIF5B-ALK	KIF5B	unknown	three_prime	29448327	19	1058	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		KIF5B (NM_004521) - ALK (NM_004304) Fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.:32305750) | Protein fusion: mid-exon (KIF5B-ALK)	NA
EVT-P-0067512-T01-IM7-297	P-0067512-T01-IM7		NOTCH1-ALK	NOTCH1	in-frame	three_prime	29449152	19	1023	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NOTCH1 (NM_017617) - ALK (NM_004304) fusion: t(2;9)(p23.2;q34.3)(chr2:g.29449152::chr9:g.139416748) | Protein Fusion: in frame {NOTCH1:ALK}	NA
EVT-P-0034044-T01-IM6-298	P-0034044-T01-IM6		NRP2-ALK	NRP2	in-frame	three_prime	29448550	19	1023	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		NRP2 (NM_201266) - ALK (NM_004304) rearrangement: c.1291+575_c.3068-119inv | Protein Fusion: in frame {NRP2:ALK}	NA
EVT-P-0055667-T06-IM7-299	P-0055667-T06-IM7		PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) | Protein Fusion: mid-exon {PICALM:ALK}	NA

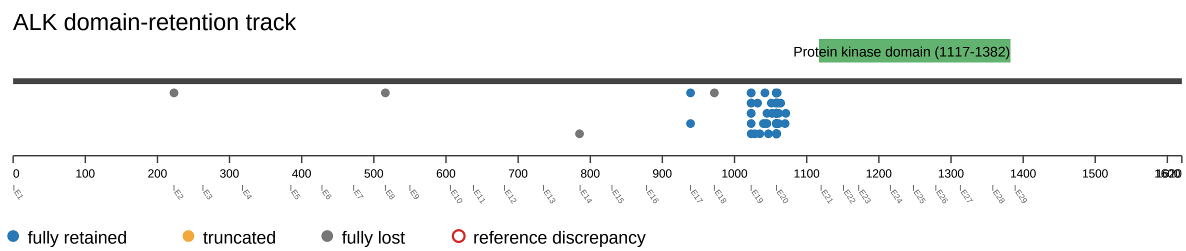
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055667-T02-IM6-300	P-0055667-T02-IM6		PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		PICALM (NM_007166) - ALK (NM_004304) rearrangement: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) | Protein Fusion: mid-exon {PICALM:ALK}	NA
EVT-P-0055667-T04-IM6-301	P-0055667-T04-IM6		PICALM-ALK	PICALM	unknown	three_prime	29446386	20	1061	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) | Protein Fusion: mid-exon {PICALM:ALK}	NA
EVT-P-0054253-T01-IM6-302	P-0054253-T01-IM6		PLEKHA7-ALK	PLEKHA7	unknown	three_prime	29448396	19	1035	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		PLEKHA7 (NM_175058) - ALK (NM_004304) fusion: t(2;11)(p23.2;p15.1)(chr2:g.29448396::chr11:g.16813809) | Protein Fusion: mid-exon {PLEKHA7:ALK}	NA
EVT-P-0059884-T01-IM7-303	P-0059884-T01-IM7		PPP1CB-ALK	PPP1CB	in-frame	three_prime	29447441	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		PPP1CB (NM_206876) - ALK (NM_004304) fusion: c.52+5708:PPP1CB_c.3172+886:ALKinv | Protein Fusion: in frame {PPP1CB:ALK}	NA
EVT-P-0048893-T01-IM6-307	P-0048893-T01-IM6		SPTBN1-ALK	SPTBN1	unknown	three_prime	29448346	19	1051	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.763+1517:SPTBN1_c.3153:ALKinv | Protein Fusion: mid-exon {SPTBN1:ALK}	NA
EVT-P-0053963-T02-IM6-308	P-0053963-T02-IM6		SPTBN1-ALK	SPTBN1	unknown	three_prime	29448345	19	1052	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.764-1123:SPTBN1_c.3154:ALKinv | Protein Fusion: mid-exon {SPTBN1:ALK}	NA
EVT-P-0045724-T01-IM6-309	P-0045724-T01-IM6		SPTBN1-ALK	SPTBN1	in-frame	three_prime	29446796	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.764-456:SPTBN1_c.3173-402:ALKinv | Protein Fusion: in frame {SPTBN1:ALK}	NA
EVT-P-0042596-T02-IM6-310	P-0042596-T02-IM6		SQSTM1-ALK	SQSTM1	in-frame	three_prime	29446511	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		SQSTM1 (NM_003900) - ALK (NM_004304) fusion: t(2;5)(p23.2;q35.3)(chr2:g.29446511::chr5:g.179254603) | Protein Fusion: in frame {SQSTM1:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042596-T01-IM6-311	P-0042596-T01-IM6		SQSTM1-ALK	SQSTM1	in-frame	three_prime	29446511	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		SQSTM1 (NM_003900) - ALK (NM_004304) fusion: t(2;5)(p23.2;q35.3)(chr2:g.29446511::chr5:g.179254603) | Protein Fusion: in frame {SQSTM1:ALK}	NA
EVT-P-0053901-T01-IM6-312	P-0053901-T01-IM6		STRN-ALK	STRN	unknown	three_prime	29446375	20	1064	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		STRN (NM_003162) - ALK (NM_004304) fusion: c.339-2565:STRN_c.3192:ALKdel | Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0048102-T03-IM6-313	P-0048102-T03-IM6		STRN-ALK	STRN	in-frame	three_prime	29447056	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		STRN (NM_003162) - ALK (NM_004304) fusion: c.412+2761:STRN_c.3173-662:ALKdel | Protein Fusion: in frame {STRN:ALK}	NA
EVT-P-0033066-T01-IM6-314	P-0033066-T01-IM6		STRN-ALK	STRN	unknown	three_prime	29448365	19	1045	False	retained	1.0	False	Protein kinase domain	PF00629; PF12810		STRN (NM_003162) - ALK (NM_004304) fusion(STRN exons 1-3 fused to ALK exons 19-29): c.413-1084:STRN_c.3134:ALKdel | Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0048102-T02-IM6-315	P-0048102-T02-IM6		STRN-ALK	STRN	in-frame	three_prime	29447056	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		STRN (NM_003162) - ALK (NM_004304) fusion: c.412+2761:STRN_c.3173-662:ALKdel | Protein Fusion: in frame {STRN:ALK}	NA
EVT-P-0056097-T01-IM6-316	P-0056097-T01-IM6		TFG-ALK	TFG	in-frame	three_prime	29447896	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		TFG (NM_001195478) - ALK (NM_004304) rearrangement: t(2;3)(p23.2;q12.2)(chr2:g.29447896::chr3:g.100456163) | Protein Fusion: in frame {TFG:ALK}	NA
EVT-P-0045363-T02-IM6-318	P-0045363-T02-IM6		TPM1-ALK	TPM1	in-frame	three_prime	29446514	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		TPM1 (NM_001018020) - ALK (NM_004304) fusion: t(2;15)(p23.2;q22.2)(chr2:g.29446514::chr15:g.63358833) | Protein Fusion: in frame {TPM1:ALK}	NA
EVT-P-0045363-T01-IM6-319	P-0045363-T01-IM6		TPM1-ALK	TPM1	in-frame	three_prime	29446514	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		TPM1 (NM_001018020) - ALK (NM_004304) fusion: t(2;15)(p23.2;q22.2)(chr2:g.29446514::chr15:g.63358833) | Protein Fusion: in frame {TPM1:ALK}	NA

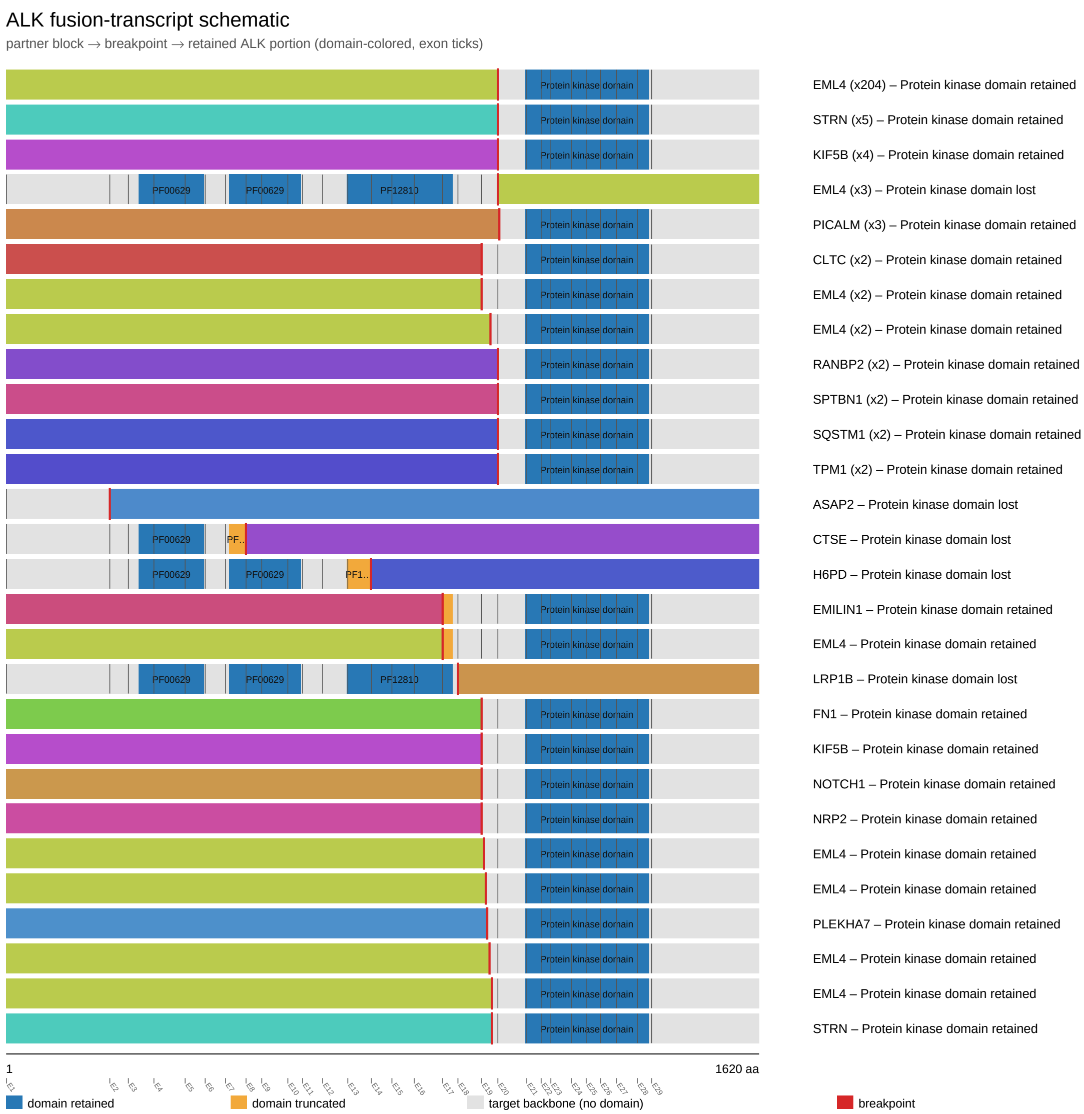
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0053383-T01-IM6-320	P-0053383-T01-IM6		TRAF3-ALK	TRAF3	in-frame	three_prime	29447688	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		TRAF3 (NM_003300) - ALK (NM_004304) fusion: t(2;14)(p23.2;q32.32)(chr2:g.29447688::chr14:g.103370282) | Protein Fusion: in frame {TRAF3:ALK}	NA
EVT-P-0001737-T01-IM3-321	P-0001737-T01-IM3		VCL-ALK	VCL	in-frame	three_prime	29447390	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		VCL (NM_014000) - ALK (NM_004304) Translocation : t(10;2)(q22.2;p23.2)(chr10:c.2435-486::chr2:c.3172+937) | Protein fusion: in frame (VCL-ALK)	NA
EVT-P-0045566-T01-IM6-322	P-0045566-T01-IM6		ZFPM2-ALK	ZFPM2	in-frame	three_prime	29447902	20	1058	True	retained	1.0	False	Protein kinase domain	PF00629; PF12810		ZFPM2 (NM_012082) - ALK (NM_004304) fusion: t(2;8)(p23.2;q23.1)(chr2:g.29447902::chr8:g.106384092) | Protein Fusion: in frame {ZFPM2:ALK}	NA

Figures

domain retention outliers



fusion schematic

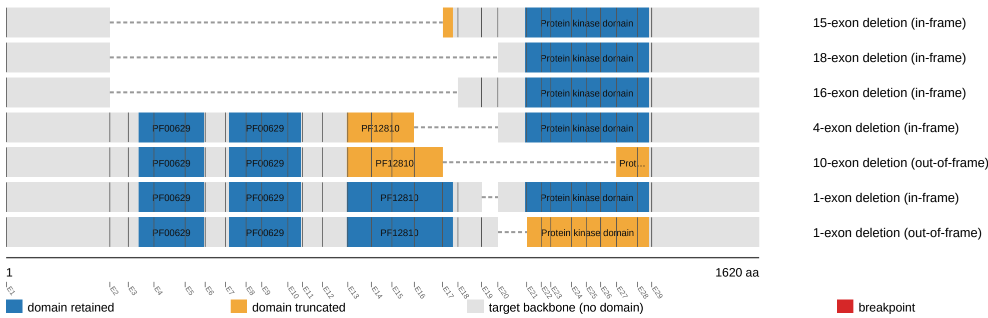


Showing the top 28 of 50 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic

ALK intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



reference comparison

Reference vs msk_impact_50k_2026